Annex E: Evaluation Rubric

Xtremis AI Challenges — Vanderbilt Dry Dock 2026

Applies to both Challenge 1 (Bloomberg Terminal for Spectrum) and Challenge 2 (The Quantum Routing Brain). Where criteria differ by challenge, both versions are given.

TECHNICAL SUBSTANCE (35%)

Does the work hold up to scrutiny? This can take many forms: a working prototype, a rigorous methodology paper with reproducible analysis, a validated framework, or a strong research synthesis with defensible projections. Whatever you build, it needs to survive hard questions. Where there are gaps due to limited time, you need defensible arguments for why it would work with more time and capital. A presentation that lacks a demo but demonstrates extremely strong research, diligent planning, and clear technical reasoning will be respected.

You do not need to use our provided annexes, data sources, or starter code. But you need to justify your scope and your data choices. If you synthesized data using AI tools, demonstrate that it is comparable to real sources and explain why the synthetic data gives you a valid basis for your conclusions.

Creative divergence from the directions and threads we provided is welcome. We will not penalize it. We will not reward it for its own sake either. Just make it work and make it good.

Challenge 1: Spectrum

A working pricing API on real public data. A methodology paper validated against auction benchmarks. A prototype that prices one band in one city with rigor. A broad framework with validated assumptions. Any of these paths is valid. Depth on a narrow slice and breadth across a wide framework are both acceptable as long as the execution is strong.

Challenge 2: Quantum

The mission threads are grounding scenarios to make the computational language concrete. Do not get lost trying to understand RF sensing workflows or orbital mechanics at the expense of building the routing layer.

Two valid paths to a strong demo:

- A working broker that routes tasks between classical and quantum backends using a free-tier quantum SDK (IBM Open Plan, AWS Braket, Qiskit Aer) on a low-qubit configuration. Extrapolate the runtime analysis against published quantum hardware roadmaps to project what the next 1–8 years look like for this problem.
- An entirely classical implementation with genuine quantitative analysis of where quantum plugs in, at what problem sizes, on what timeline, and what speedup is expected based on published benchmarks.

Overclaiming quantum advantage will hurt you. Honest framing of “classical today, quantum when the hardware catches up, and here is the math on when that is” will help you.

VENTURE STRATEGY (25%)

This is not a pitch deck exercise. We are not looking for a TAM number on a slide. We want to see that you have thought about how to actually build a company around this problem over the next 12–24 months.

Questions we want you to address (not all, but the ones relevant to your approach):

- What distinguishes your prototype from an MVP from a product? What are the gates between each stage?
- What are the capital requirements at each gate? Where does non-dilutive financing fit? These are deep-tech problems with natural pathways through SBIR, STTR, NSF, DARPA, and other government research funding. Show us you know those pathways exist.

- Who are your first design partners? Name the type of organization (not necessarily a specific company) that would co-develop with you and why.
- How do you recruit domain experts? Spectrum economics, quantum computing, defense systems—these are hard hiring markets. What is your strategy?
- How do you maintain equity while growing? How do you think about dilution, bootstrapping with government contracts, and strategic partnerships?
- What does a credible go-to-market look like? Not "sell to carriers" or "sell to DoD." What is the specific entry point, the first contract, the wedge?

PRESENTATION & NARRATIVE (15%)

Can you hold a room? Is the thesis clear in 60 seconds? Can you explain your technical choices to a non-specialist without losing accuracy? Can you answer hard questions without retreating to buzzwords?

We will ask you technical questions about your domain. For Challenge 1, expect questions about spectrum regulation, auction mechanics, sharing frameworks, and pricing methodology. For Challenge 2, expect questions about quantum computing fundamentals, tactical compute architectures, classification constraints, and the honest state of quantum advantage. The purpose is not to quiz you on trivia. It is to test whether you understand the problem deeply enough to have built something meaningful, or whether you outsourced the thinking to an AI and wrapped it in a slide deck.

ENGINEERING QUALITY (15%)

Six days is not long and we do not expect production code. But we expect sustainable, auditable practices:

- Clean repository structure with a readable README.
- Code that a new team member could pick up and understand.
- Containerization where appropriate (Docker Compose is sufficient).
- Evidence that AI-generated code was reviewed, understood, and tested rather than pasted blindly. Agentic coding tools are great and there are clever ways to ensure quality. We are looking for the clever ways.
- Version control with meaningful commit history (not one giant commit at the end).

This matters because it affects your ability to scale, survive audits, onboard collaborators, and attract engineering talent. It is a signal of how you would run a real company.

TEAM EXECUTION & INDIVIDUAL ACCOUNTABILITY (10%)

This criterion scores whether the team functions like a real founding team under pressure.

- Did you divide work intelligently across the team?
- Can each person speak with authority on their piece and defend it under questioning?
- Did anyone carry dead weight? A team with strong technical output where one person did everything and four people watched is a 0/10 on this criterion. A team where everyone owns a clear piece is a 10/10.
- Can you articulate your own contribution without hedging?

Alongside scoring, we collect background context on each team member for our own evaluation of potential hires/investments. This is not scored but is requested:

- Your academic standing (e.g., CS junior, ECE graduate student).
- Something that genuinely motivates you: a research experience, a class that changed how you think, an internship, a club you built, a project you are proud of, a movie, book, or hobby that excites you. We want to understand what drives you, not read a resume.